

Name:

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Roll No:

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- Write your name cleanly in **CAPITAL** letters and roll number inside the boxes.
- Write answers in the space provided only. Maximum marks: 25 Time: 18:30 - 19:45.

1. Suppose that $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ is continuous and Lipschitz in its second variable, i.e. $|F(r, s) - F(r, t)| \leq K|s - t|$ for all $r \in \mathbb{R}$. Define $T : C([0, 1]) \rightarrow C([0, 1])$ by $(Tf)(x) = \int_0^x F(t, f(t)) dt$.

- (a) Justify whether T is always a strict contraction or not.
 (b) Show that T maps bounded sets into equicontinuous sets.

$F : \mathbb{R}^2 \rightarrow \mathbb{R}$ is continuous.
 and Lipschitz in its second variable.

$$|F(r, s) - F(r, t)| \leq K|s - t|$$

$$T : C([0, 1]) \rightarrow C([0, 1])$$

$$(Tf)(x) = \int_0^x F(t, f(t)) dt.$$

(a). for T to be strict contraction.

$$\|Tf - Tg\|_{\infty} \leq K \|f - g\|_{\infty}$$

$$\text{now } \left\| \int_0^x F(t, f(t)) dt - \int_0^x F(t, g(t)) dt \right\|$$

$$\leq \left\| \int_0^x F(t, f(t)) - F(t, g(t)) dt \right\|$$

$$\leq \int_0^x \sup_{t \in [0, 1]} |F(t, f(t)) - F(t, g(t))| dt$$

$$\leq \sup_{t \in [0, 1]} \int_0^x |F(t, f(t)) - F(t, g(t))| dt$$

$$\leq \sup_{t \in [0, 1]} \int_0^x K |f(t) - g(t)| dt.$$

$$\leq K \sup_{t \in [0, 1]} \int_0^x |f(t) - g(t)| dt \quad [3+4]$$

$$= K \|f - g\|_{\infty}$$

now this depends on value of K and for strict contraction $0 < K < 1$ which is not given. hence it is not always a strict contraction.

(b) Let $A \subset C([0, 1])$ that is bounded.

$$\exists M \text{ s.t. } \|f\|_{\infty} < M \quad \forall f \in A.$$

now consider $g \in T(A)$

$$\text{where } g = T(f) = \int_0^x F(t, f(t)) dt$$

for equicontinuous. take $\epsilon > 0 \exists \delta > 0$

$$|g(y) - g(x)| < \epsilon \text{ when } |y - x| < \delta$$

$$\text{now } |g(y) - g(x)| = \left| \int_x^y F(t, f(t)) dt \right|$$

w. log assume $y > x$

$$|g(y) - g(x)| = \left| \int_x^y F(t, f(t)) dt \right|$$

$$\leq \int_x^y |F(t, f(t))| dt$$

$$\leq \int_x^y M dt = M(y - x)$$

now we know $|f(t)| < M \quad \forall t \in [0, 1]$

$$\Rightarrow (t, f(t)) \in [0, 1] \times [-M, M]$$

this is a compact set in $\mathbb{R}^2$. $\exists F(t, f(t))$ attains a maximum denote it by L .

$$\leq \int_x^y L dt = L(y - x)$$

now

$$|g(y) - g(x)| \leq L|y - x|$$

$$\text{take } \delta = \frac{\epsilon}{L} \quad \delta = \frac{\epsilon}{L}$$

$$\text{let } |y - x| < \delta$$

$$\text{then } |g(y) - g(x)| \leq L|y - x|$$

$$\leq L \frac{\epsilon}{L}$$

$$= \epsilon$$

$$\Rightarrow |g(y) - g(x)| \leq \epsilon \text{ when } |y - x| < \delta$$

be independent of g & y, x

hence equicontinuous

..... Rough Work

$\exists f_n$. f_n (there are polynomials that $f_n(x_m)$)

$$\|f_n - g\|$$

to be equicontinuous.

$$|g(y) - g(x)|$$

$$< \epsilon \text{ whenever.}$$

2. Let $\{x_i\}_{i \in \mathbb{N}} \subset (0, 1)$ be a sequence such that $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n x_i^k$ exists for every $k \in \mathbb{N}$. Show that

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{i=1}^n \frac{1}{2 + \sin(x_i)} = 0.$$

$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n x_i^k$ exists for all k .

define $I(f) := \frac{1}{n} \sum_{i=1}^n f(x_i)$

take polynomial $p = \sum_{j=0}^m a_j x^j$

$$\text{now } I(p) = \frac{1}{n} \sum_{i=1}^n \sum_{j=0}^m a_j x_i^j$$

$$\lim_{n \rightarrow \infty} I(p) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \sum_{j=0}^m a_j x_i^j = \sum_{j=0}^m a_j \left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n x_i^j \right)$$

[allowed since finite terms]

$$\text{now } \lim_{n \rightarrow \infty} I(p) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \sum_{j=0}^m a_j x_i^j$$

$$= \sum_{j=0}^m a_j \left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n x_i^j \right)$$

[allowed since finite no of terms]

$$\lim_{n \rightarrow \infty} I(p) = \lim_{n \rightarrow \infty} \sum_{j=0}^m a_j \left(\frac{1}{n} \sum_{i=1}^n x_i^j \right)$$

$$= \sum_{j=0}^m a_j \lim_{n \rightarrow \infty} \left(\frac{1}{n} \sum_{i=1}^n x_i^j \right)$$

$$= \sum_{j=0}^m a_j L_j$$

hence the limit of $I(p)$ exists

for all polynomials.

take $f = \frac{1}{2 + \sin(x)}$ (it is continuous)

* take $\epsilon > 0$ polynomial p_m s.t. $\|f - p_m\|_\infty < \frac{\epsilon}{n}$

(Weierstrass approximation)

$$I(f) - I(p_m) = \frac{1}{n} \sum_{i=1}^n \left(\frac{1}{2 + \sin(x_i)} - p_m(x_i) \right)$$

$$\text{now } f(x_i) - p_m(x_i) < \frac{\epsilon}{n}$$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^n \left(\frac{1}{2 + \sin(x_i)} - p_m(x_i) \right) < \frac{1}{n} \sum_{i=1}^n \frac{\epsilon}{n}$$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^n \frac{1}{2 + \sin(x_i)} < \frac{1}{n} \sum_{i=1}^n \frac{\epsilon}{n} + \frac{1}{n} \sum_{i=1}^n p_m(x_i)$$

now also

now as $p_m \rightarrow f$ (uniformly)

$$\frac{1}{n} \sum_{i=1}^n p_m(x_i) \rightarrow \frac{1}{n} \sum_{i=1}^n f(x_i)$$

$$= \frac{1}{n} \sum_{i=1}^n \lim_{m \rightarrow \infty} p_m(x_i)$$

$$= \lim_{m \rightarrow \infty} \left(\frac{1}{n} \sum_{i=1}^n p_m(x_i) \right)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n f(x_i) = \lim_{n \rightarrow \infty} \lim_{m \rightarrow \infty} \left(\frac{1}{n} \sum_{i=1}^n p_m(x_i) \right)$$

$$= \lim_{m \rightarrow \infty} \left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n p_m(x_i) \right)$$

limit swap on uniform convergence

$$= \lim_{m \rightarrow \infty} L_m = L \quad (\text{hence limit exists})$$

and is finite

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \left(\sum_{i=1}^n f(x_i) \right) =$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \cdot \lim_{m \rightarrow \infty} \left(\frac{1}{n} \sum_{i=1}^n f(x_i) \right) = 0 \cdot L = 0$$

3. Let $\{g_n\}_{n \in \mathbb{N}} \subset C([a, b])$ and for each $n \in \mathbb{N}$ we have $g_n(x) \geq -\frac{1}{n^2}$ for every $x \in [a, b]$. If $\sum_{n=1}^{\infty} g_n$ converges pointwise to a continuous function on $[a, b]$, show that $\sum_{n=1}^{\infty} g_n$ converges uniformly on $[a, b]$.

[3]

$$\{g_n\}_{n \in \mathbb{N}} \subset C([a, b])$$

$$\{g_n\}_{n \in \mathbb{N}} \subset C([a, b])$$

$$g_n(x) \geq -\frac{1}{n^2} \quad \forall x \in [a, b]$$

$$\sum_{n=1}^{\infty} g_n \text{ converges pointwise to } g.$$

$$\text{i.e. } g(x) = \sum_{n=1}^{\infty} g_n(x)$$

$$\text{take } S_n(x) = \sum_{i=1}^n g_i(x)$$

$$S_n(x) \rightarrow S(x) \text{ ptwise on } [a, b]$$

$$\Rightarrow \forall \epsilon > 0 \quad \exists N_{\epsilon, x} \text{ s.t.}$$

$$|S_n(x) - S(x)| < \epsilon$$

$$\forall n \geq N_{\epsilon, x}$$

as